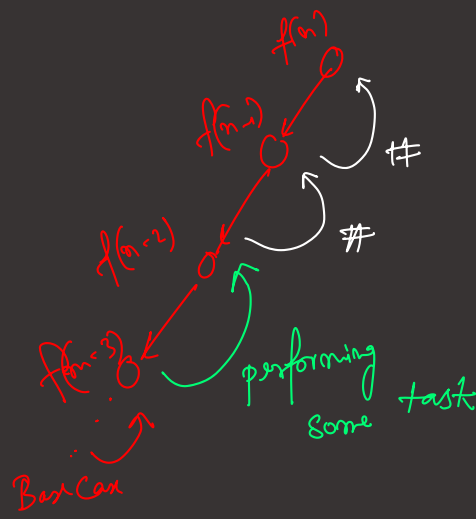
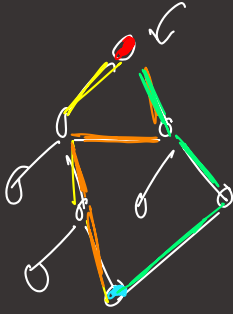


Backtracking

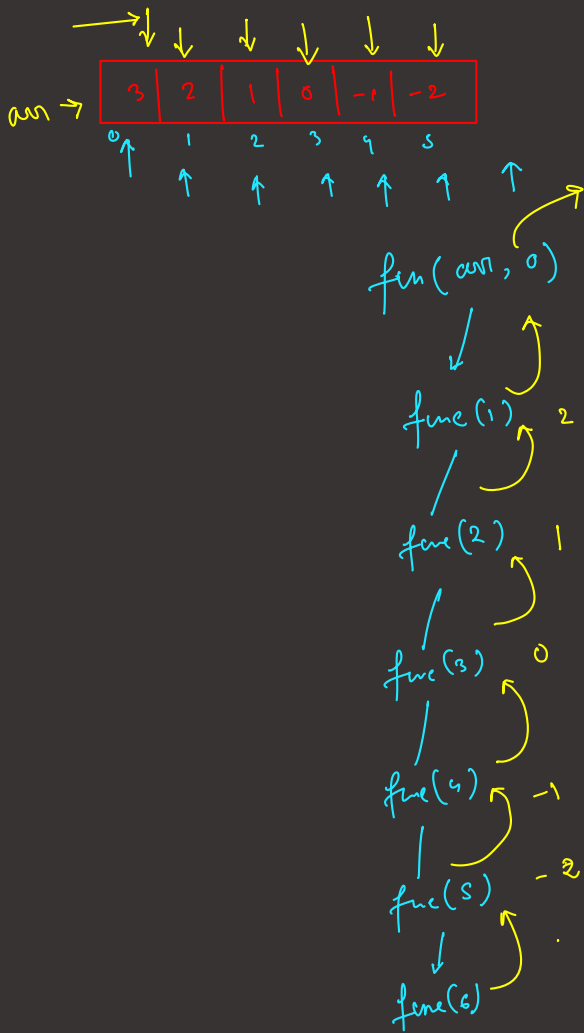


types - ~~1~~ All possible ways.

② Best way among all the solⁿ *

✓ any one way (it may not be best)

③ Decision → whether the solⁿ exist or not.



```

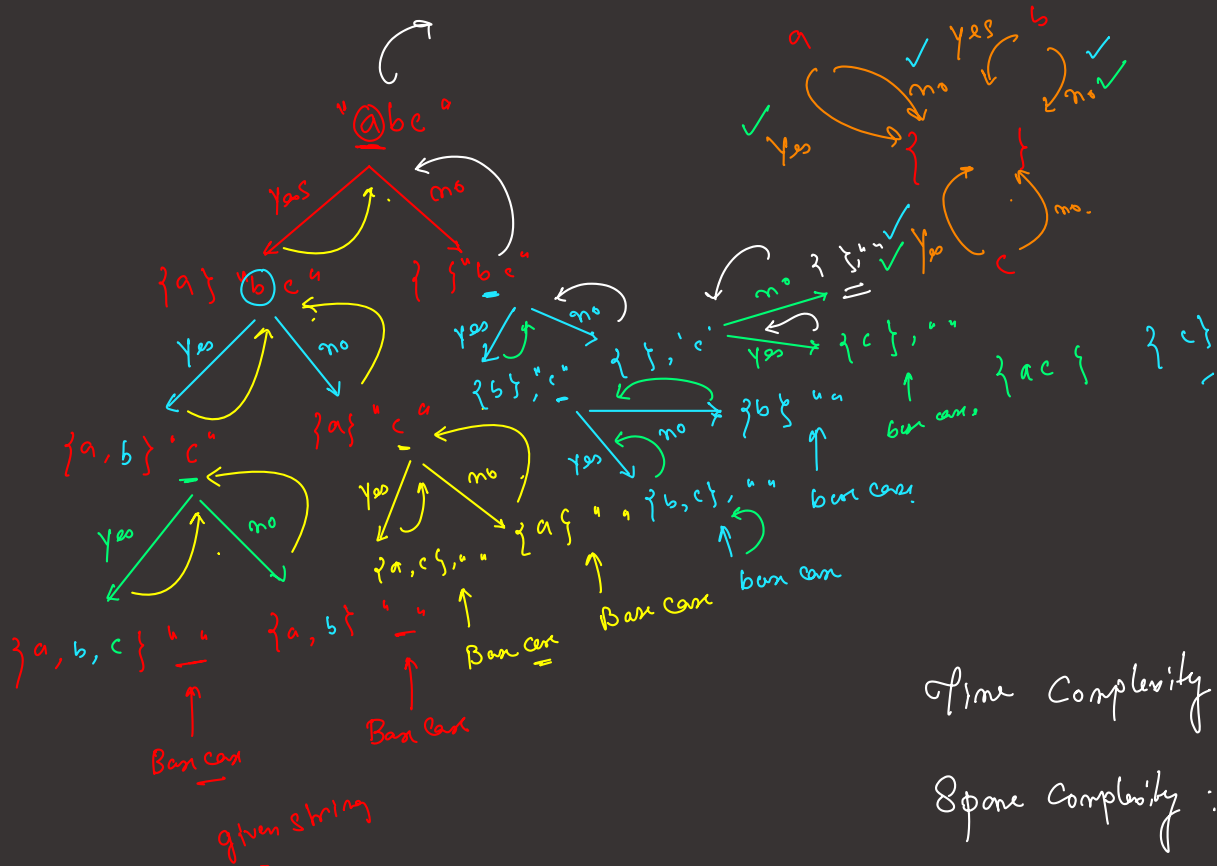
fun(arr, i) {
    if (i == arr.length) {
        return;
    }
    fun(arr, i+1);
    s.out(arr[i]);
}
    
```

o/p: -2 -1 0 1 2 3

Q Find and print all subsets of a given string

eg. str = "abc"

o/p: abc, ab, bc, ca, a, b, c, ""



Time Complexity: $O(2^n \times n)$

Space Complexity: $O(n)$

given string
findSubsets(set, ans, i)

if (i == set.length()) {

print(ans);

return;

}

// recursion

// yes

findSubsets(set, ans + set.charAt(i), i + 1);

// no

findSubsets(set, ans, i + 1);

}

f(abc, "", 0)

↓

f(abc, 'a', 1)

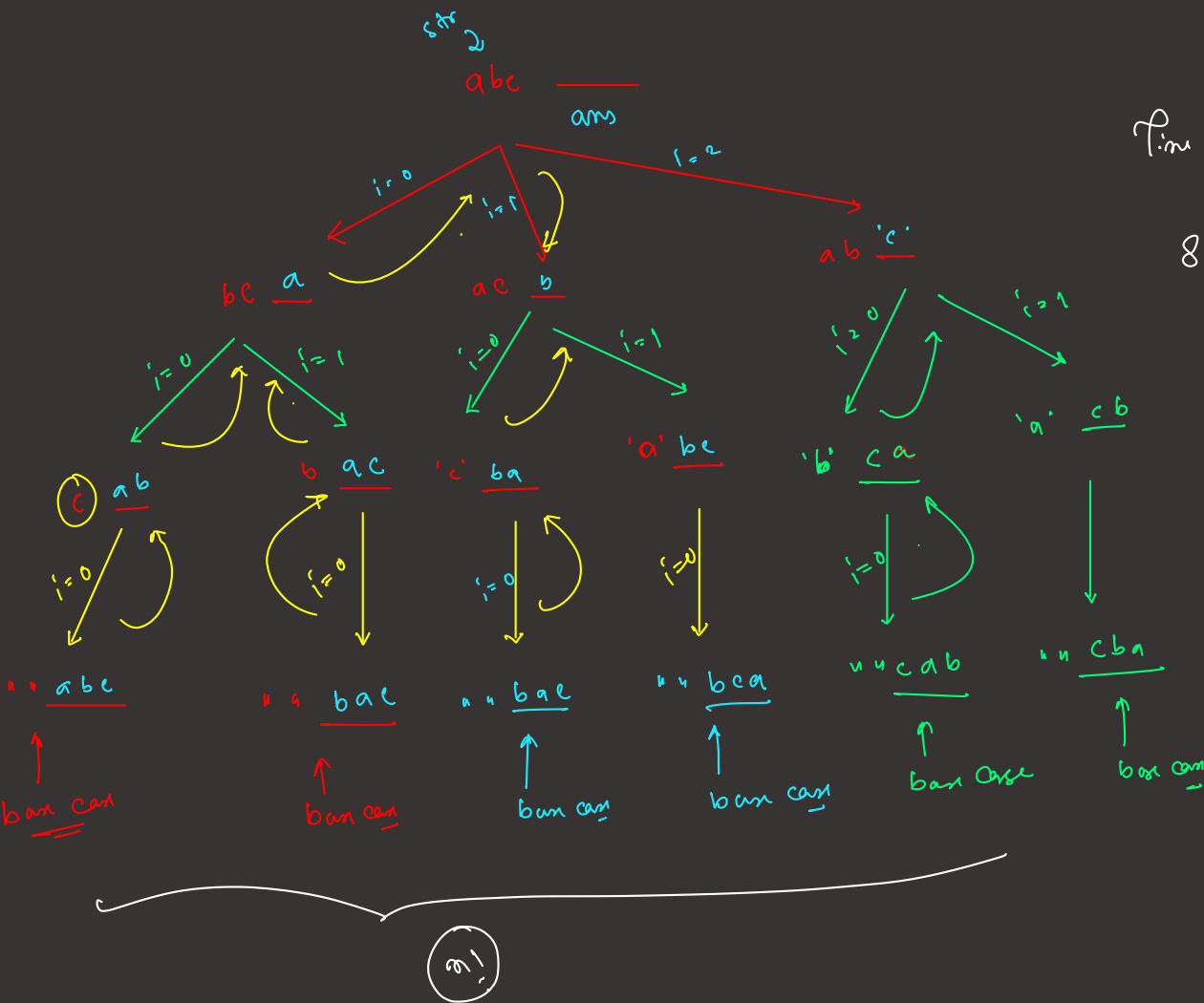
f(abc, 'b', 2)

Q Find Permutation

Print all the permutation of a string.

i/p: "abc"

o/p: "abc", "acb", "bac", "bca", "cab", "cba"



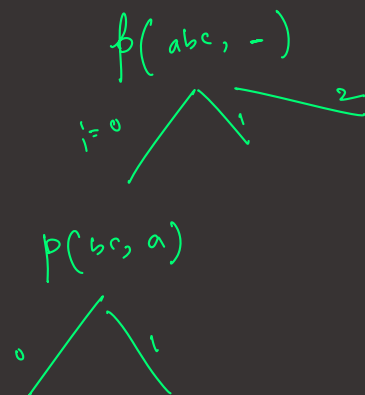
$\uparrow$ (b) c
 $\uparrow$ ac
 $\uparrow$ abc
 $\uparrow$ abc + ef
 $\uparrow$ str, str.substr(0, i)
 $\uparrow$ str, str.substr(i+1, ...)

Space

$p(abc, "")$
generate?

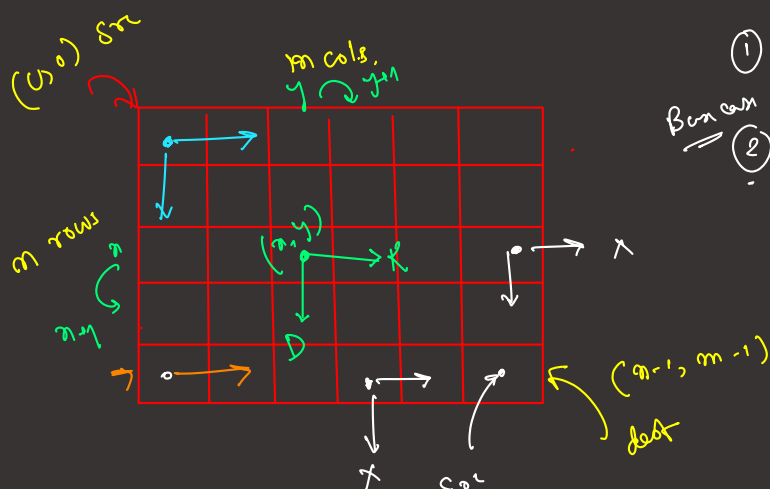
o/p: abc, acb, bac, bca, ...

Time

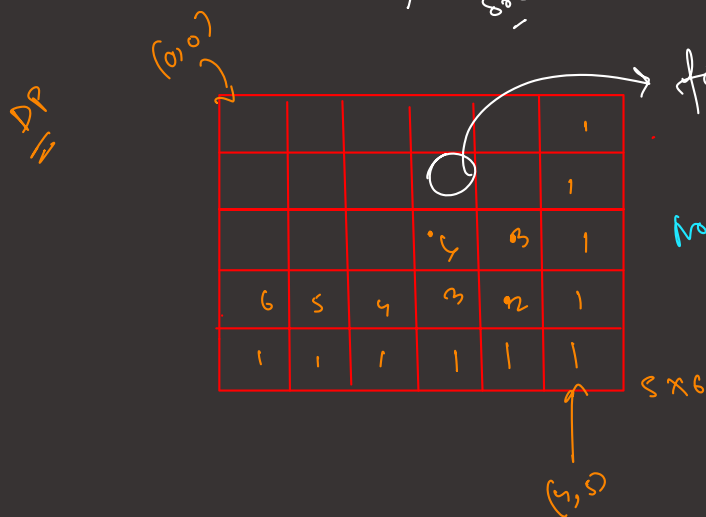


Day-20

Grid ways: Find the no. of ways to reach from $(0,0)$ to $(n-1, m-1)$ in an $n \times m$ grid.
Allowed Moves - right and down.

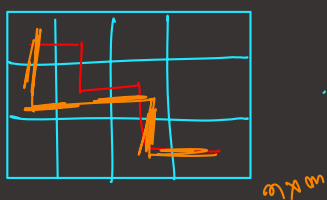


- ① $ways(n, y) = ways(n, y+1) + ways(n+1, y)$
- ② $src == dest \rightarrow ways = 1$
- ③ for edges $\rightarrow ways = 1$.



for a particular cell = $(m+n)$ ways to reach dest
now for each cell there is 2 ways
 $\therefore$ total ways = $2^{(m+n)}$

$$T.C = O(2^{m+n})$$



Right turn = $(m-1)$ R
Down turn = $(n-1)$ D

Arrangement of this R, D

$$\frac{(m+n-2)!}{(m-1)!(n-1)!}$$

$$T.C = O(m+n)$$

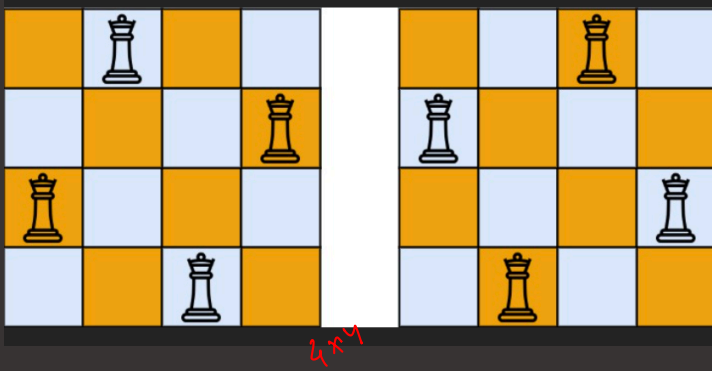
factorial $f_n \rightarrow$ recursion

$$\left. \begin{array}{c} fact(n) \\ \downarrow \\ fact(n-1) \\ \vdots \\ fact(1) \end{array} \right\} O(n)$$

DRRDR
RDRDR

N Queen Problem

placed N queen on $N \times N$ chess-board such that no 2 queen can attack each other

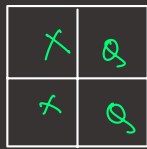
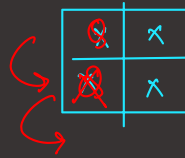
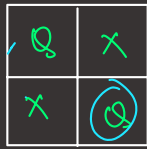


No attack

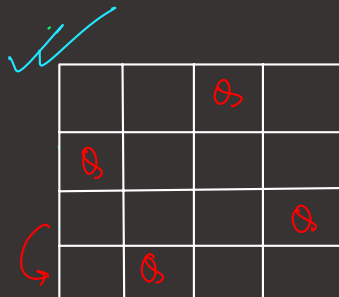
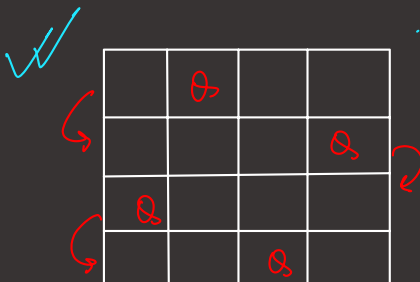
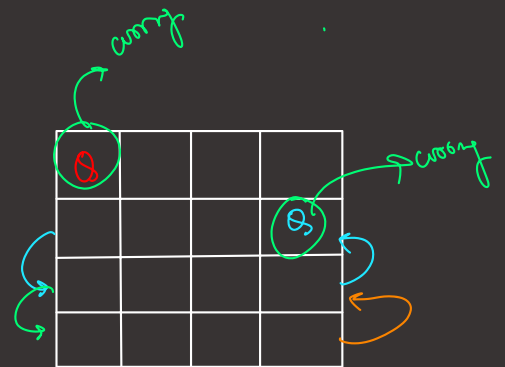
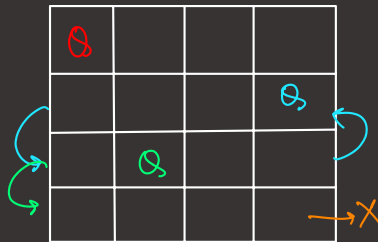
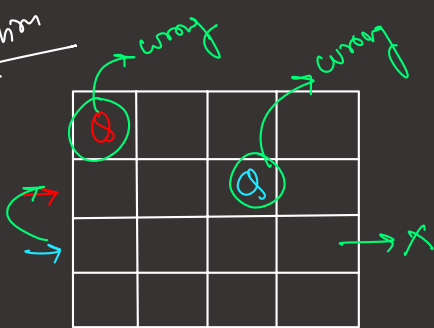
1 row = 1 Queen

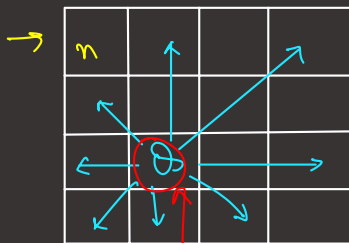
row

N=2

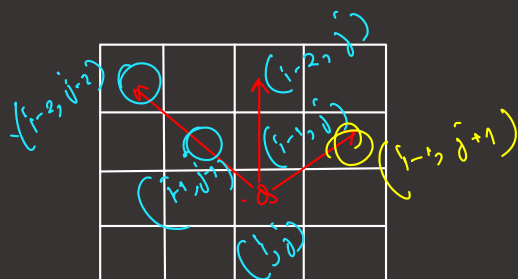


Resolution





position is safe or not



safe $\rightarrow$ pop $\rightarrow$ backtrack (no & there)
 $\rightarrow$ L. Diag. $\rightarrow$ no attack
 $\rightarrow$ R. diagonal $\rightarrow$ no attack

Time Complexity: $O(n^n) \rightarrow O(n!)$

$$T(n) = (\text{Queen Place time}) T(n-1) + \underbrace{is\ safe}_{O(n)}$$

$\downarrow$
 $O(n)$

$$T(n) = n T(n-1) + n$$

$$T(n) = n T(n-1) + n$$

$$T(n-1) = (n-1) T(n-2) + (n-1)$$

$$T(n-2) = (n-2) T(n-3) + (n-2)$$

$\vdots$

$$T(1) = T(0) + 1$$

$$\begin{aligned} T(n) &= n \left\{ (n-1) T(n-2) + (n-1) \right\} + n \\ &= n! + n! + n^2 \\ &= \underline{\underline{O(n!)}} \end{aligned}$$

Q Count total no. of $\{N\text{-Queen Sol}^n\}$

H.W $N\text{-knight Problem}$